

4. Project Planning Phase

1. INTRODUCTION

The Project Planning Phase divides the overall application lifecycle of **ComicCraft** into structured, manageable milestones and tactical activities. This document outlines the roadmap across model selection and architecture design, core module functionality, application routing via `routes.py`, frontend interface development, and cloud deployment.

2. OBJECTIVES

The primary purpose of Phase 4 is to establish a rigorous operational framework for development. Key objectives include:

- Define concrete project milestones and breakdown tasks for implementation.
- Establish a clear sequential implementation order and phase dependencies.
- Plan core module and cross-component architectural dependencies.
- Structure GitHub repository management through standardized issues and milestones.
- Prepare mechanisms for local deployment, verification, and automated testing.

3. PHASE WORKFLOW

The lifecycle proceeds sequentially through key developmental targets designed to ensure continuous integration and seamless feature delivery:

Milestone	Focus Area & Description
1. Model & Architecture	Selecting ML/AI model pipelines and defining system architectural contracts.
2. Core Functionality	Developing background processors, backend engines, and core generation logic.
3. Application Routes	Integrating backend endpoints and requests orchestration inside <code>routes.py</code> .
4. Frontend UI/UX	Building user interfaces, canvas interactivities, and responsive components.
5. Deployment	Configuring local staging setups, containerization, and production deployment.

4. DETAILED EXPLANATION

Effective planning establishes a strict sequence of execution to prevent architectural friction. The project begins with initial requirements gathering and system design, establishing solid data models and interface definitions. Subsequently, core logic modules are implemented and verified in isolation. Once core stability is reached, API routes and HTML/CSS template integration proceed via `routes.py`. Finally, end-to-end user verification and staging deployment ensure production readiness.

5. GITHUB-READY CODE / STRUCTURE

To streamline team coordination and issue tracking on GitHub, standardized phase labels are established:

Label Name	Phase Description
phase-1: ideation	Brainstorming, scope formulation, and conceptualization
phase-2: requirements	Functional specifications and user story definition
phase-3: design	UI/UX wireframes, architecture, and database schemas
phase-4: planning	Task breakdown, scheduling, and milestone setup
phase-5: development	Core feature coding and integration implementation
phase-6: testing	Unit testing, integration tests, and QA verification
phase-7: documentation	API docs, user guides, and technical specifications
phase-8: demonstration	Final showcase, presentation builds, and release review

6. GITHUB FILES & DELIVERABLES

Key Project Artifacts:

- `docs/project-plan.md` — Master project plan document tracking design and deployment strategies.
- **GitHub Issues** — Granular feature requests and bug reports tagged by phase.
- **GitHub Milestones** — Grouped deliverables enforcing project timelines.
- **Commit History** — Traceable, semantic git commit sequence validating execution.

7. EXPECTED OUTCOME

At the conclusion of Phase 4, all planning aspects of the **ComicCraft** application are clearly documented, structured, and verified according to project guidelines. Specific completion metrics and phase validation rely on the student's repository configuration and verified commit history.

8. PHASE CONCLUSION

Phase 4 provides a robust, structured foundation for the upcoming development stages of **ComicCraft**. By establishing clear milestones, task dependencies, and standardizing GitHub project workflows, the project maintains clarity, predictability, and efficiency across evaluation and continuous execution.